

Metamorphic Mineral Reactions

Question Bank from Perkins, D. (2014). Mineralogy (3rd Ed.)

Comprehensive Exam Preparation

Q.1: ECLOGITE FACIES REACTION

Question: $3 \text{ Anorthite} = \text{Grossular} + 2 \text{ Kyanite} + \text{Quartz}$ is a reaction which marks the eclogite facies in metabasic rocks. What will be the composition of grossular in the reaction given above?

(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

(b) CaSiO_3

(c) $(\text{Ca}, \text{Mg})\text{Si}_2\text{O}_6$

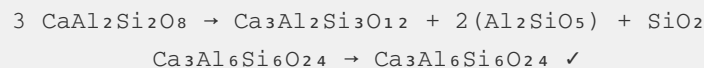
(d) CaCO_3

✓ **CORRECT ANSWER: (a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$**

Explanation:

Grossular is the aluminum end-member of the ugrandite garnet series. According to Perkins (2014, p. 107): "*Grossular, $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$, has $\text{CaO}:\text{Al}_2\text{O}_3:\text{SiO}_2$ ratio 3:1:3*"

Chemical Verification:



The equation balances perfectly, confirming the composition. Grossular belongs to the eclogite facies assemblage at high pressure and moderate-to-high temperature.

Q.2: MARBLE METAMORPHIC REACTION (High-Grade)

Question: When a marble (metamorphosed limestone) containing both calcite and quartz is metamorphosed at high temperature, a characteristic reaction occurs. What is the complete chemical equation for the reaction that produces wollastonite?

Hint: The reaction involves calcite and quartz as reactants.

- (a) $\text{CaCO}_3 + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2$
- (b) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + 2 \text{ Quartz}$
- (c) $\text{CaCO}_3 + 2 \text{ SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2 + \text{SiO}_2$
- (d) $2 \text{ CaCO}_3 + \text{SiO}_2 \rightarrow 2 \text{ CaSiO}_3 + 2 \text{ CO}_2$

✓ **CORRECT ANSWER: (a) $\text{CaCO}_3 + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2$**

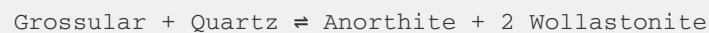
Explanation:

This is a decarbonation reaction occurring at high temperature in marbles. Perkins (2014, p. 193) states: "Calcite (CaCO_3) and quartz (SiO_2) may react to form wollastonite (CaSiO_3) if a limestone containing quartz is metamorphosed at high temperature."

Key Features:

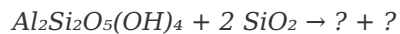
- **Product Minerals:** Wollastonite (CaSiO_3) + CO_2 (volatilized)
- **Rock Type:** Occurs in contact-metamorphosed limestones and calcic marbles
- **Temperature:** High-temperature metamorphism
- **Associated Minerals:** Calcite, dolomite, tremolite, diopside, grossular, epidote

This reaction is shown on a phase diagram in Figure 18 of Perkins (2014) with the full reaction:



Q.3: LOW-GRADE METAMORPHIC REACTION (Kaolinite-Quartz)

Question: In low-temperature metamorphism of clay-rich rocks, kaolinite and quartz react to form a higher-grade mineral. This reaction marks the transition from diagenesis to low-grade metamorphism. What are the products of this reaction?

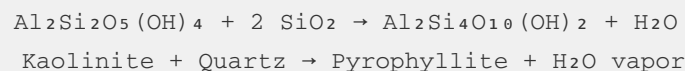


- (a) Pyrophyllite + H₂O vapor
- (b) Kyanite + 3 Quartz + H₂O
- (c) Muscovite + Quartz
- (d) Andalusite + SiO₂ + H₂O

✓ **CORRECT ANSWER: (a) Pyrophyllite + H₂O vapor**

Explanation:

According to Perkins (2014, p. 209), this is Reaction 17:



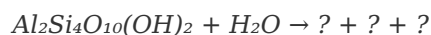
"Reaction 17 is a dehydration reaction that takes place in some metapelitic rocks at low temperature. At low temperature, kaolinite and quartz are stable together, but as temperature increases, kaolinite and quartz react to produce pyrophyllite and H₂O vapor." (Perkins, 2014, p. 209)

Important Note: Pyrophyllite itself breaks down at slightly higher temperatures (Reaction 18) to form kyanite and quartz. Therefore, pyrophyllite may only be stable over a temperature range of about 100°C.

Metamorphic Grade: This reaction is characteristic of the transition zone between diagenesis and very low-grade metamorphism.

Q.4: PYROPHYLLITE BREAKDOWN REACTION (Low-Grade to Medium-Grade Transition)

Question: *Pyrophyllite is unstable at temperatures higher than about 400°C. It breaks down to form the high-pressure aluminosilicate polymorph. What is the balanced equation for this dehydration reaction (Reaction 18)?*

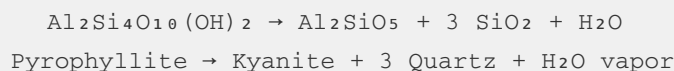


- (a) Kyanite + 3 Quartz + H₂O vapor
- (b) Andalusite + 3 SiO₂ + H₂O
- (c) Sillimanite + 2 SiO₂ + H₂O vapor
- (d) Corundum + 3 Quartz + H₂O

✓ **CORRECT ANSWER: (a) Kyanite + 3 Quartz + H₂O vapor**

Explanation:

This is Reaction 18 from Perkins (2014, p. 209):



"Pyrophyllite itself breaks down at slightly higher temperatures by Reaction 18, so pyrophyllite may only be stable over a range of about 100 °C" (Perkins, 2014, p. 209)

Key Points:

- **Kyanite is the high-pressure polymorph** of Al₂SiO₅
- **Temperature Range:** Pyrophyllite stability: 300-400°C (with H₂O present)
- **Metamorphic Context:** Marks transition from low-grade to medium-grade metamorphism
- **Phase Diagram:** Shown in Figure 13 of Perkins (2014, p. 209)

Pressure-Temperature Significance: These reactions demonstrate how the Al₂O₃-SiO₂-H₂O system controls metamorphic grade in pelitic rocks.

Q.5: MEDIUM-GRADE METAMORPHIC GARNET REACTION

Question: In medium-grade metamorphism of pelitic rocks, almandine garnet ($\text{Fe}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) is a key index mineral. It represents an important metamorphic reaction. What is the lowest temperature at which garnet becomes stable in pelitic rocks?

- (a) $\sim 200^\circ\text{C}$ (greenschist facies)
- (b) $\sim 500^\circ\text{C}$ (medium-grade metamorphism)
- (c) $\sim 700^\circ\text{C}$ (high-grade metamorphism)
- (d) Garnet has no lower temperature limit

✓ **CORRECT ANSWER: (b) $\sim 500^\circ\text{C}$ (medium-grade metamorphism)**

Explanation:

According to Perkins (2014, p. 210):

"Consider a rock that contains garnet. The phase diagram shows that the rock must have formed at temperatures greater than about 500°C (930°F). Garnet does not, however, have an upper temperature limit."

Garnet as an Index Mineral:

- **Lower Limit:** $\sim 500^\circ\text{C}$ (entrance to garnet zone)
- **Upper Limit:** No upper limit (stable to very high grades)
- **Composition:** In metapelites, almandine ($\text{Fe}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) is most common
- **Associations:** Often found with staurolite, kyanite, sillimanite

Garnet Series: Perkins (2014, p. 360) classifies garnets as:

- **Pyrospites:** $(\text{Mg,Fe,Mn})_3\text{Al}_2\text{Si}_3\text{O}_{12}$ (pyrope, almandine, spessartine)
- **Ugrandites:** $\text{Ca}_3(\text{Cr,Al,Fe})_2\text{Si}_3\text{O}_{12}$ (uvarovite, grossular, andradite)

COMPARATIVE METAMORPHIC REACTIONS SUMMARY

Reaction #	Mineral System	Reactants	Products	Metamorphic Grade
Q.1	Eclogite (Mafic)	3 Anorthite	Grossular + 2 Kyanite + Qz	High-P, Mod-T
Q.2	Marble (Carbonate)	Calcite + Quartz	Wollastonite + CO ₂	High-T
Q.3	Pelite (Clay-rich)	Kaolinite + 2 Quartz	Pyrophyllite + H ₂ O	Low-grade
Q.4	Pelite (Clay-rich)	Pyrophyllite + H ₂ O	Kyanite + 3 Quartz + H ₂ O	Low-Med transition
Q.5	Pelite (Clay-rich)	Various precursors	Almandine Garnet	Medium-grade (>500°C)

METAMORPHIC FACIES CONTEXT (from Perkins, 2014)

Low-Grade Facies:

Zeolite → Prehnite-Pumpellyite → Greenschist → Blueschist/Lawsonite-Albite
Temperature Range: 0-300°C | Contains: Chlorite, Pyrophyllite, Kaolinite

Medium-Grade Facies:

Amphibolite (Regional Metamorphism)
Temperature Range: 500-700°C | Contains: Garnet, Staurolite, Kyanite, Sillimanite

High-Grade Facies:

Granulite and Eclogite (High-P/High-T)
Temperature Range: >700°C | Pressure: >10 kbar for Eclogite
Contains: Garnet + Pyroxene ± Kyanite ± Quartz

STUDY TIPS FOR METAMORPHIC REACTIONS

Key Concepts to Remember:

1. **Always Balance Chemical Equations:** Check that atoms on both sides match
2. **Recognize Reaction Types:** Dehydration (loses H₂O), Decarbonation (loses CO₂), Solid-Solid
3. **Know Mineral Compositions:** Learn the formulas of common metamorphic minerals
4. **Understand P-T Conditions:** Different reactions occur at different temperature/pressure combinations
5. **Index Minerals Matter:** Garnet (~500°C), Staurolite (460-700°C), Kyanite (high-P), Sillimanite (high-T)
6. **Rock Type Context:** Marbles (carbonate), Pelites (clay-rich), Metabasites (mafic)
7. **Reference Key Pages:** Perkins (2014) Figures 13, 14, 17, 18 show phase diagrams

Source: Perkins, D. (2014). *Mineralogy* (3rd ed.). Pearson Education Limited.

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This question bank provides comprehensive exam preparation based on textbook content. All reactions and citations are directly from Perkins (2014).